

Causal Effects in Policy - Lecture 2

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- Prove the Law of Iterated Expectations (LIE)
- Conditional Expectation Function (CEF) - Definition & 3 Properties
- Best Linear Predictor as an approximation to CEF
- Regressions with Discrete Covariates (Dummies)

Law of Iterated Expectations

- (X, Y) are two r.v.'s with joint pdf $f(X, Y)$
 - Marginal densities: $f(X) = \int_Y f(X, Y)dY$, and conditional density:
 $f(Y|X) = \frac{f(X, Y)}{f(X)}$.
- Expectation: $E[Y] = \int_Y f(Y)dY$, conditional
 $E[Y|X] = \int_Y Yf(Y|X)dY$, which is a function of X .
- Law of iterated expectations (LIE):

$$E[Y] = E[E[Y|X]] \quad (1)$$

Proof (continuous r.v.'s):

$$\begin{aligned} E[E[Y|X]] &= \int_X E[Y|X]f(X)dx = \int_X \left(\int_Y Yf(Y|X)dY \right) f(X)dX \\ &= \int_Y Y \left(\int_X f(Y|X)f(X)dX \right) dY \\ &= \int_Y Yf(Y) dY = E[Y] \end{aligned}$$

Conditional Expectation Function

- CEF $E[Y|X]$ is a function of X that represents the expectation of Y conditional on random variables X .
- Let $\epsilon = Y - E[Y|X]$. *Two excellent properties of CEF:*
 - ④ $E[\epsilon] = 0$, $E[\epsilon|X] = 0$ and $E[\epsilon h(X)] = 0$ for *any* function of X .

Proof:

$$E[\epsilon] = E[Y] - E[E[Y|X]] = 0 \text{ by LIE} \quad (2)$$

$$E[\epsilon|X] = E[Y|X] - E[E[Y|X]|X] = E[Y|X] - E[Y|X] = 0$$

Remark: $E[\epsilon|X] = 0$ means ϵ is mean independent of X . With $E[\epsilon|X] = 0$, by LIE,

$$E[\epsilon h(X)] = E[E[\epsilon h(X)|X]] = E[E[\epsilon|X] h(X)] = 0 \quad (3)$$

Conditional Expectation Function

- ② the function $E[Y|X]$ minimizes $E[(Y - m(X))^2]$

Proof: given any function m of r.v. X ,

$$\begin{aligned} Y - m(X) &= Y - E[Y|X] + E[Y|X] - m(X) \\ \Rightarrow (Y - m(X))^2 &= (Y - E[Y|X])^2 + (E[Y|X] - m(X))^2 \\ &\quad + 2(Y - E[Y|X]) \underbrace{(E[Y|X] - m(X))}_{\text{funx of } X: h} \\ \Rightarrow E[(Y - m(X))^2] &= E[\epsilon^2] + \underbrace{E[(E[Y|X] - m(X))^2]}_{\geq 0} + \underbrace{2E[\epsilon h(X)]}_{=0} \end{aligned}$$

The minimizing choice is $m(X) = E[Y|X]$!

So: we've established that among the functions of X , the CEF $E[Y|X]$ gives the "best guess" for Y in the sense of minimizing $E[(Y - m(X))^2]$.

Linear Approximation to CEF

- *Problem:* we don't know the conditional density $f(Y|X)$.
- Solution: we'll use the "linear regression function": combination $x_i\beta$.
- Recall: given a sample of size N the OLS regression coefficients β solve:

$$\min_{\beta} \sum_{i=1}^N (y_i - x_i'\beta)^2$$

- Consider WLLN for the r.v. $(y_i - x_i'\beta)^2$: we have $\frac{1}{N} \sum_{i=1}^N (y_i - x_i'\beta)^2 \rightarrow^p E[(y_i - x_i'\beta)^2]$
- The "infeasible" (or population) OLS estimator solves:

$$\begin{aligned} \min_{\beta} E[(y_i - x_i'\beta)^2] \\ \Rightarrow \beta^* = E[x_i x_i']^{-1} E[x_i y_i] \end{aligned} \tag{4}$$

This gives us the best linear predictor of Y given X : $E^*[y_i|x_i] = x_i'\beta^*$ (proof below).

Linear Approximation to CEF

How does $x_i' \beta^*$ relate to $E[y_i | x_i]$?

- 1 If the CEF is linear $E[y_i | x_i] = x_i' \beta^e$, then $\beta^* = \beta^e$.

Why? Recall that if we define the CEF error $\varepsilon_i = y_i - E[y_i | x_i]$,

$$E[x_i \varepsilon_i] = 0 \Rightarrow E[x_i (y_i - x_i' \beta^e)] = 0 \Rightarrow \beta^e = \beta^*$$

This means that *if the true CEF is linear*, then the infeasible OLS represents the CEF.

- This happens when x_i' s are dummies since $E[y_i | x_i]$ is $E[y_i | i \text{ in group } k]$

Linear Approximation to CEF

- ② $E^*[y_i|x_i] = x_i'\beta^*$ is the “best” linear approx. to $E[y_i|x_i]$ (best as in *minimum-MSE*)

$$\beta^* = \operatorname{argmin}_{\beta} E[(E[y_i|x_i] - x_i'\beta)^2] \quad (5)$$

Proof:

$$\begin{aligned} y_i - x_i'\beta &= y_i - E[y_i|x_i] + E[y_i|x_i] - x_i'\beta \\ \Rightarrow E[(y_i - x_i'\beta)^2] &= E[\epsilon_i^2] + E[(E[y_i|x_i] - x_i'\beta)^2] \\ &\quad + 2 \underbrace{E[\epsilon_i (E[y_i|x_i] - x_i'\beta)]}_{=0} \end{aligned}$$

$$E[(E[y_i|x_i] - x_i'\beta)^2] = E[(y_i - x_i'\beta)^2] - E[\epsilon_i^2]$$

So $x_i'\beta^*$ that minimizes the mean squared error also minimizes $E[(E[y_i|x_i] - x_i'\beta)^2]$.

We can think of the “population regression” as:

$$y_i = x_i'\beta^* + u_i$$

which satisfies $E[x_i u_i] = 0$ (why?). Unless the CEF is linear, we won't have $E[u_i|x_i] = 0$.

- We have established that $E^*[y_i|x_i] = x_i'\beta^*$ is the best linear approximation to the CEF $E[y_i|x_i]$.

$$\begin{aligned}\beta^* &= \operatorname{argmin}_{\beta} E[(y_i - x_i'\beta)^2] \\ &= E[x_i x_i']^{-1} E[x_i y_i]\end{aligned}$$

- How do find an estimate for β^* ? The feasible OLS minimizes the sum of squared residuals (SSR) in the sample:

$$\hat{\beta} = \operatorname{argmin}_{\beta} \sum_{i=1}^N (y_i - x_i'\beta)^2 \quad (6)$$

The FOC (in vector form) are:

$$\sum_{i=1}^N -2x_i(y_i - x_i'\beta) = 0$$

which implies that

$$\begin{aligned}\sum_{i=1}^N x_i x_i' \beta &= \sum_{i=1}^N x_i y_i \\ \Rightarrow \hat{\beta} &= \left[\sum_{i=1}^N x_i x_i' \right]^{-1} \sum_{i=1}^N x_i y_i \\ &= \left[\frac{1}{N} \sum_{i=1}^N x_i x_i' \right]^{-1} \left(\frac{1}{N} \sum_{i=1}^N x_i y_i \right)\end{aligned}$$

Now plug in the population regression: $y_i = x_i' \beta^* + u_i$, where $E[u_i x_i] = 0$,

$$\begin{aligned}\hat{\beta} &= \left[\frac{1}{N} \sum_{i=1}^N x_i x_i' \right]^{-1} \frac{1}{N} \sum_{i=1}^N x_i (x_i' \beta^* + u_i) \\ &= \beta^* + \left[\frac{1}{N} \sum_{i=1}^N x_i x_i' \right]^{-1} \left[\frac{1}{N} \sum_{i=1}^N x_i u_i \right]\end{aligned}\tag{7}$$

Feasible OLS

- Consistency: $\hat{\beta} \xrightarrow{P} \beta^*$ by WLLN and Slutsky Theorem.
- Asymptotic distribution:
 - by the Central Limit ThM and $E[x_i u_i] = 0$:

$$\sqrt{N} \left(\frac{1}{N} \sum_{i=1}^N x_i u_i - E[x_i u_i] \right) \rightarrow^d N(0, E[x_i x_i' u_i^2])$$

note the variance of K -by-1 vector $x_i u_i$ (u_i is a scalar): $\text{var}(x_i u_i) = E[x_i u_i u_i x_i'] - E[x_i u_i] E[x_i u_i]' = E[x_i x_i' u_i^2] - 0_{K \times K} = E[x_i x_i' u_i^2]$.

- By Slutsky ThM, we have:

$$\sqrt{N}(\hat{\beta} - \beta^*) \rightarrow^d N(0, E[x_i x_i']^{-1} E[x_i x_i' u_i^2] E[x_i x_i']^{-1}) \quad (8)$$

when do we have $E[x_i x_i' u_i^2] = E[x_i x_i'] \times \text{var}(u_i)$? Apply LIE:

$$\begin{aligned} E[x_i x_i' u_i^2] &= E_x[E[x_i x_i' u_i^2 | x]] \\ &= E_x[x_i x_i' E[u_i^2 | x]] \end{aligned}$$

Homoskedasticity: $E[u_i^2 | x] = E[u_i^2]$. Otherwise the variance of residual vary by x . We need to compute robust heteroskedasticity-robust standard errors.

When is CEF linear? $E[y|x] = E^*[y|x]$

- Recall in the linear normal model, we assume:

$$y_i = \beta_0 + \beta_1 x_i + u_i, u_i \sim N(0, \sigma^2) \quad (9)$$

in which the normally distributed U is independent of X :
 $E[u_i|x_i] = 0$. This is a special case where the CEF is linear:

$$E[y_i|1, x_i] = \beta_0 + \beta_1 x_i$$

note the constant 1 is often merged into x_i :

$$E[y_i|x_i] = x_i' \beta = \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

- Another important case: regressions with discrete regressors! x_i are indicators/dummies...

Regressions with Dummy Variables

- Consider a (population) regression with a binary independent variable $D_i \in \{0, 1\}$:

$$\begin{aligned}y_i &= \alpha + \beta_1 D_i + \epsilon_i \\E[\epsilon_i] &= E[\epsilon_i D_i] = 0\end{aligned}\tag{10}$$

Recall this represents the best linear projection of y on D ,
 $E^*[y_i|D_i] = \alpha + \beta_1 D_i$.

$$\begin{aligned}(\alpha, \beta_1) &= \underset{(a,b)}{\operatorname{argmin}} E[(y_i - a - bD_i)^2] \\ \alpha &= E[y_i] - \beta_1 E[D_i] \\ 0 &= E[(y_i - \alpha - \beta_1 D_i)D_i] = E[\epsilon_i D_i] \\ \rightarrow \beta_1 &= \frac{\operatorname{cov}(y_i, D_i)}{\operatorname{var}(D_i)} \text{ shown in lecture 1 (univariate reg)}\end{aligned}$$

Regressions with Dummy Variables

- Since D_i is a dummy variable/indicator, we have $E[D_i^2] = E[D_i] = \Pr(D_i = 1)$. Denote by $p = E[D_i]$.

$$\begin{aligned} E[y_i D_i] &= E_{D_i}[E[y_i D_i | D_i]] \\ &= p \times E[y_i D_i | D_i = 1] + (1 - p) \times \underbrace{E[y_i D_i | D_i = 0]}_0 \\ &= p \times E[y_i | D_i = 1] \end{aligned}$$

therefore,

$$\begin{aligned} \text{cov}(y_i, D_i) &= E[y_i D_i] - E[y_i]E[D_i] = E[y_i | D_i = 1] - E[y_i] * p \\ \text{var}(D_i) &= E[D_i^2] - E[D_i]^2 = p(1 - p) \end{aligned}$$

- prove:

$$\begin{aligned} \beta_1 &= E[y_i | D_i = 1] - E[y_i | D_i = 0] \\ \alpha &= E[y_i | D_i = 0] \end{aligned} \tag{11}$$

Regressions with Dummy Variables

- In summary, with a single binary variable, the best linear projection of y_i on $(1, D_i)$ is:

$$\begin{aligned} E^*[y_i|D_i] &= E[y_i|D_i = 0] + D_i \times (E[y_i|D_i = 1] - E[y_i|D_i = 0]) \\ &= E[y_i|D_i] \text{ CEF!!} \end{aligned}$$

we have shown that in this case the conditional expectation of any outcome y_i on a discrete variable is Linear! The constant is the mean of the outcome among $D_i = 0$, and the slope is the difference in means for the two groups.

Regressions with Dummy Variables

- In a randomized control trial, we have 2 groups: the treatment group, with $D_i = 1$, and the control group, with $D_i = 0$. So we could fit a regression model to the “pooled” data

$$y_i = \alpha + \beta D_i + \epsilon_i$$

The treatment effect of interest is $E[y_i|D_i = 1] - E[y_i|D_i = 0]$. (We will discuss this in the potential outcome framework.)

- The (feasible) OLS estimator: let $\bar{Y}_0$ denote the sample mean of the outcome for the control group, and $\bar{Y}_1$ for the treatment group,

$$\hat{\alpha} = \bar{Y}_0 \rightarrow^P E[y_i|D_i = 0] \tag{12}$$

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0 \rightarrow^P E[y_i|D_i = 1] - E[y_i|D_i = 0]$$

So we get the estimated mean of the control group in the intercept, and the estimated treatment effect in the coefficient of D_i .

Regressions with Multiple Dummies

- Now we generalize the result to a regression with multiple indicators/dummies.
- Suppose individuals belong to G mutually exclusive groups. For example, education level: less than HS, HS, College and above.
- Define $D_{gi} = 1[i \in g]$ for $g = 1, 2, \dots, G$. Because of the mutual exclusivity, $D_{gi} \times D_{ki} = 0$ for any $g \neq k$.
- We posit a population regression:

$$y_i = x_i' \beta + u_i \quad (13)$$

$$x_i := \begin{bmatrix} D_{1i} \\ \dots \\ D_{Gi} \end{bmatrix}$$

Given a random sample, let $N_g = \sum_i D_{gi}$ denote the num. individuals in group g . The fraction of individuals in g : $\bar{p}_g = \frac{1}{N} \sum_i D_{gi} = \frac{N_g}{N}$. Recall the OLS estimator:

$$\hat{\beta} = \left[\frac{1}{N} \sum_i x_i x_i' \right]^{-1} \frac{1}{N} \sum_i x_i y_i$$

Regressions with Multiple Dummies

- Now let's break it down:

$$\begin{aligned}\frac{1}{N} \sum_i x_i x_i' &= \begin{pmatrix} \frac{1}{N} \sum D_{1i}^2 & \frac{1}{N} \sum D_{1i} D_{2i} & \dots & \frac{1}{N} \sum D_{1i} D_{Gi} \\ \frac{1}{N} \sum D_{2i} D_{1i} & \frac{1}{N} \sum D_{2i}^2 & \dots & \\ \dots & & & \\ & & & \frac{1}{N} \sum D_{Gi}^2 \end{pmatrix} \\ &= \begin{pmatrix} \bar{p}_1 & 0 & \dots & 0 \\ 0 & \bar{p}_2 & \dots & 0 \\ \dots & & & \\ 0 & & & \bar{p}_G \end{pmatrix}\end{aligned}$$

And with

$$\frac{1}{N} \sum_{i=1}^N D_{gi} y_i = \frac{1}{N} \sum_{i \in g} y_i = \frac{N_g}{N} \times \left(\frac{1}{N_g} \sum_{i \in g} y_i \right) = \bar{p}_g \times \bar{y}_g,$$

$$\frac{1}{N} \sum_i x_i y_i = \begin{pmatrix} \frac{1}{N} \sum D_{1i} y_i \\ \frac{1}{N} \sum D_{2i} y_i \\ \dots \\ \frac{1}{N} \sum D_{Gi} y_i \end{pmatrix} = \begin{pmatrix} \bar{p}_1 \bar{y}_1 \\ \bar{p}_2 \bar{y}_2 \\ \dots \\ \bar{p}_G \bar{y}_G \end{pmatrix}$$

Regressions with Multiple Dummies

So

$$\hat{\beta} = \begin{pmatrix} \bar{p}_1 & 0 & \dots & 0 \\ 0 & \bar{p}_2 & \dots & 0 \\ \vdots & & & \\ 0 & & & \bar{p}_G \end{pmatrix}^{-1} \begin{pmatrix} \bar{p}_1 \bar{y}_1 \\ \bar{p}_2 \bar{y}_2 \\ \vdots \\ \bar{p}_G \bar{y}_G \end{pmatrix} = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_G \end{pmatrix} \quad (14)$$

Also:

$$\bar{x} = \frac{1}{N} \sum_i x_i = \begin{pmatrix} \frac{1}{N} \sum D_{1i} \\ \vdots \\ \frac{1}{N} \sum D_{Gi} \end{pmatrix} = \begin{pmatrix} \bar{p}_1 \\ \vdots \\ \bar{p}_G \end{pmatrix}$$

So obviously

$$\bar{y} = \bar{x}' \hat{\beta} = \sum_g \bar{p}_g \bar{y}_g \quad (15)$$

So the OLS regression is just a way to get the group-specific means.

Regressions with Multiple Dummies

The population version:

$$\begin{aligned} E^*[y_i|x_i] &= E[x_i]'\beta \\ &= \sum_g E[D_{gi}] \times E[y_i|D_{gi} = 1] \\ &= E[y_i|x_i] \end{aligned}$$

the linear projection is also the CEF!

Regressions with Multiple Dummies

- What if there is a constant in x_i : $x_i' = [1, D_{1i}, D_{2i}, \dots, D_{Gi}]$?
 $\sum_{g=1}^G D_{gi} = 1$ (the first element of x_i)
 - Multicollinearity: The matrix $\frac{1}{N} \sum_i x_i x_i'$ is singular (not invertible)!
the sum of columns 2-($G+1$) equals the first column (1's).

$$\frac{1}{N} \sum_i x_i x_i' = \begin{pmatrix} 1 & \bar{p}_1 & \bar{p}_2 & \dots & \bar{p}_G \\ \bar{p}_1 & \bar{p}_1 & 0 & \dots & 0 \\ \bar{p}_2 & 0 & \bar{p}_2 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ \bar{p}_G & 0 & 0 & \dots & \bar{p}_G \end{pmatrix} \quad (16)$$

note the 2nd-($G+1$)-th column sum up to the the first column (using $\sum_g \bar{p}_g = 1$). The covariates $x_i = [1, D_{1i}, D_{2i}, \dots, D_{Gi}]$ with dummies for every possible group and a constant are linearly dependent!

Regressions with Multiple Dummies

- In Stata (see reg_dummy.do), you may run “reg y ibn.region” which automatically generate $D_{1i}, D_{2i}, \dots, D_{Gi}$ for you. By default it will keep constant, but drop a group as base to ensure $\frac{1}{N} \sum_i x_i x_i'$ is nonsingular:

```
. reg tempjuly ibn.region, robust // West as base
note: 4.region omitted because of collinearity.
```

```
Linear regression                               Number of obs   =       954
                                                F(3, 950)       =     392.53
                                                Prob > F        =     0.0000
                                                R-squared       =     0.4247
                                                Root MSE       =     4.1746
```

tempjuly	Robust		t	P> t	[95% conf. interval]	
	Coefficient	std. err.				
region						
NE	1.241406	.4451532	2.79	0.005	.3678092	2.115004
N Cntrl	1.35866	.4451531	3.05	0.002	.4850625	2.232257
South	8.881006	.4468209	19.88	0.000	8.004136	9.757876
West	0	(omitted)				
_cons	72.10859	.405254	177.93	0.000	71.3133	72.90389

Regressions with Multiple Dummies

- And if you suppress constant, Stata will keep all groups and the coef on D_{gi} : $\hat{\beta}_g = \bar{y}_g$,

```
. reg tempjuly ibn.region, robust nocons
```

Linear regression

```
Number of obs   =      954
F(4, 950)       >    99999.00
Prob > F        =      0.0000
R-squared       =      0.9969
Root MSE      =      4.1746
```

tempjuly	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
region						
NE	73.35	.1842025	398.20	0.000	72.98851	73.71149
N Cntrl	73.46725	.1842024	398.84	0.000	73.10576	73.82874
South	80.9896	.1881971	430.34	0.000	80.62027	81.35893
West	72.10859	.405254	177.93	0.000	71.3133	72.90389

Regressions with Multiple Dummies

- In Python (see `reg_dummy.py`), I use `statsmodel-OLS`. By default it will keep constant, but drop a group as base to ensure $\frac{1}{N} \sum_i x_i x_i'$ is nonsingular: to be consistent with Stata, I use “West” as the base/reference group.

```
=====
Dep. Variable:      tempjuly    R-squared:      0.425
Model:              OLS        Adj. R-squared:  0.423
Method:              Least Squares    F-statistic:    392.5
Date:                Thu, 04 Sep 2025    Prob (F-statistic): 8.61e-166
Time:                10:21:30    Log-Likelihood: -2715.0
No. Observations:    954        AIC:              5438.
Df Residuals:        950        BIC:              5457.
Df Model:             3
Covariance Type:     HCL
=====
```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	72.1086	0.405	177.934	0.000	71.314	72.903
C(region, Treatment(reference='West'))[T.NE]	1.2414	0.445	2.789	0.005	0.369	2.114
C(region, Treatment(reference='West'))[T.N Cntrl]	1.3587	0.445	3.052	0.002	0.486	2.231
C(region, Treatment(reference='West'))[T.South]	8.8810	0.447	19.876	0.000	8.005	9.757

Regressions with Multiple Dummies

- Suppress the constant by adding “0+” before other covariates.
Python - smf.ols also keeps all groups and the coef on D_{gi} : $\hat{\beta}_g = \bar{y}_g$,

Dep. Variable:	tempjuly	R-squared:	0.425
Model:	OLS	Adj. R-squared:	0.423
Method:	Least Squares	F-statistic:	nan
Date:	Thu, 04 Sep 2025	Prob (F-statistic):	nan
Time:	10:21:30	Log-Likelihood:	-2715.0
No. Observations:	954	AIC:	5438.
Df Residuals:	950	BIC:	5457.
Df Model:	3		
Covariance Type:	HC1		

	coef	std err	z	P> z	[0.025	0.975]
C(region)[NE]	73.3500	0.184	398.203	0.000	72.989	73.711
C(region)[N Cntrl]	73.4673	0.184	398.840	0.000	73.106	73.828
C(region)[South]	80.9896	0.188	430.344	0.000	80.621	81.358
C(region)[West]	72.1086	0.405	177.934	0.000	71.314	72.903